

VCTK 1000-Pair Analysis

In-Distribution Evaluation Report

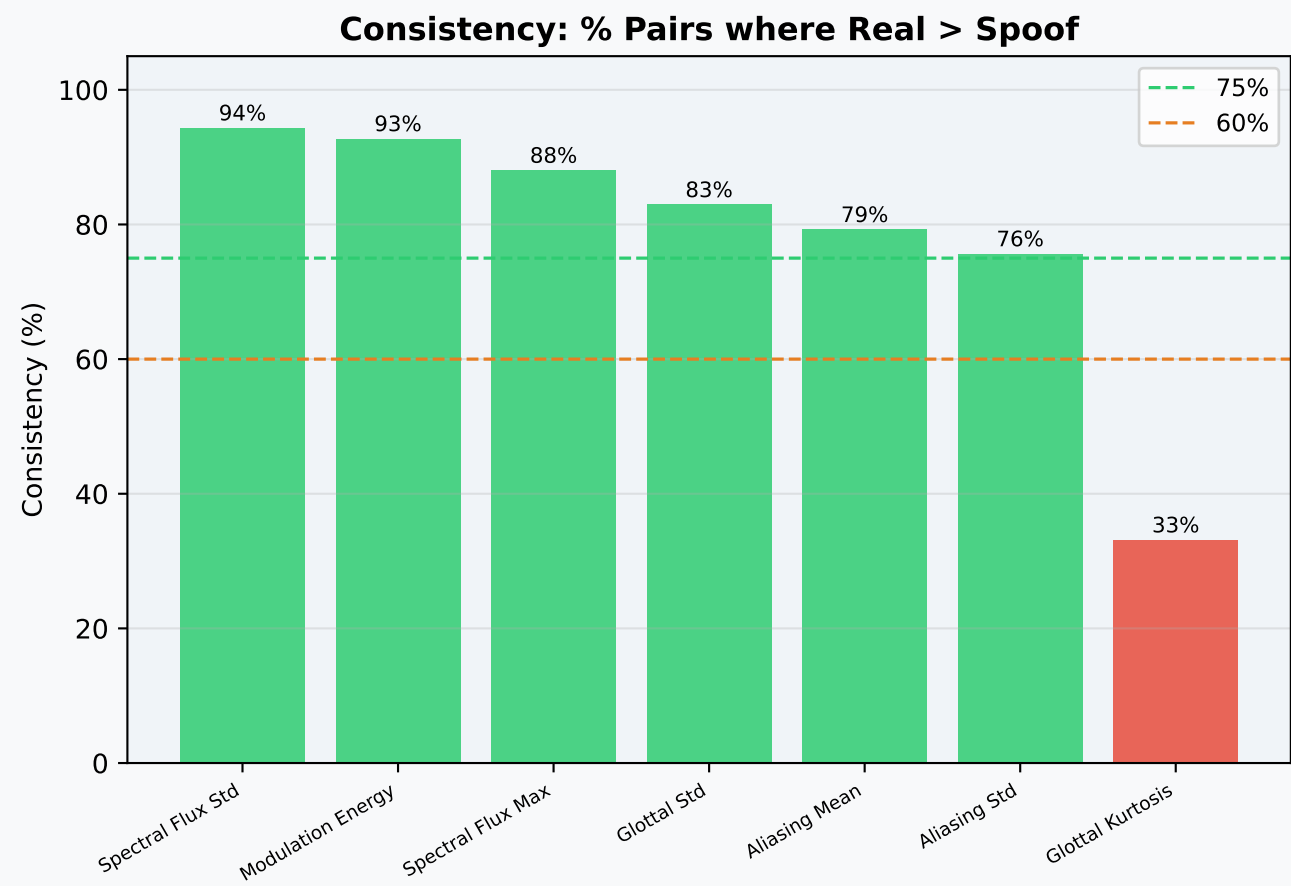
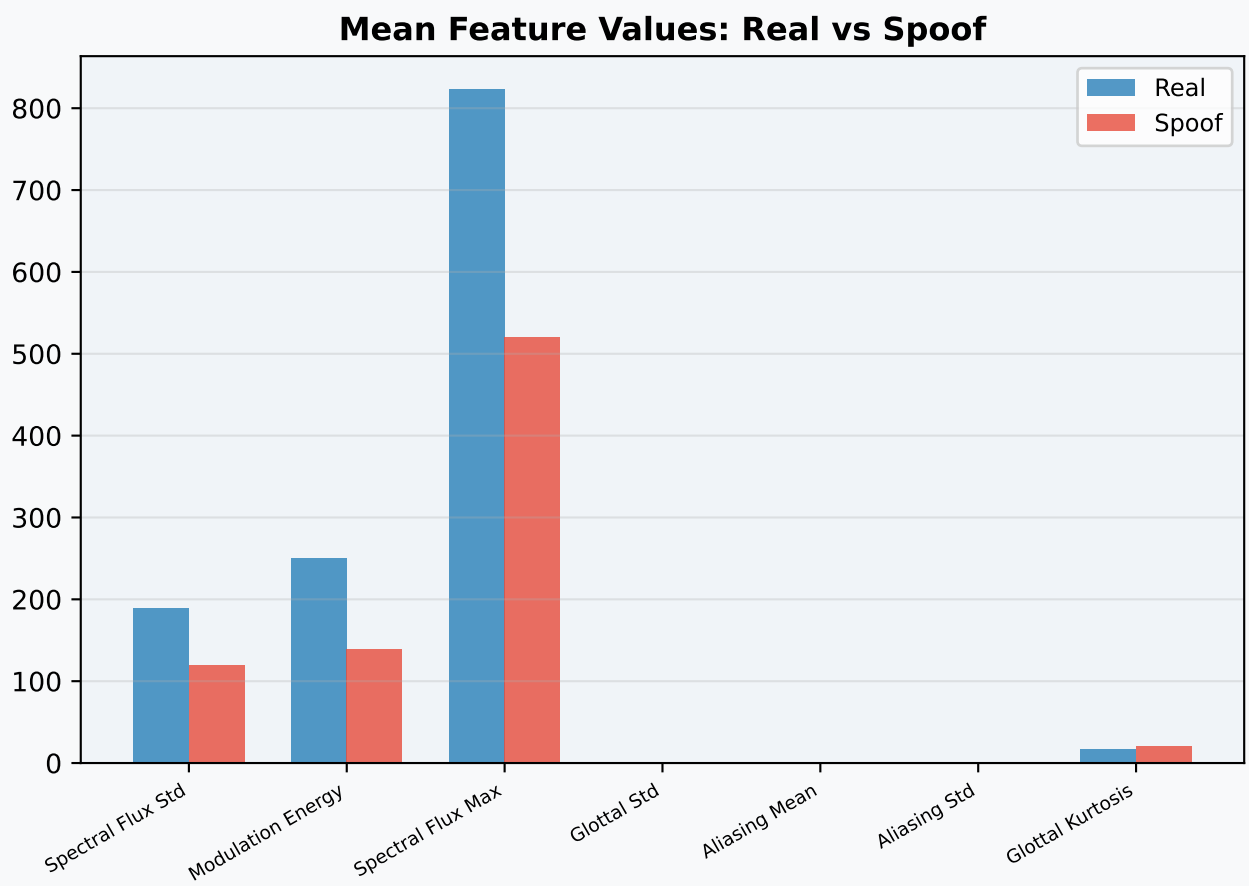
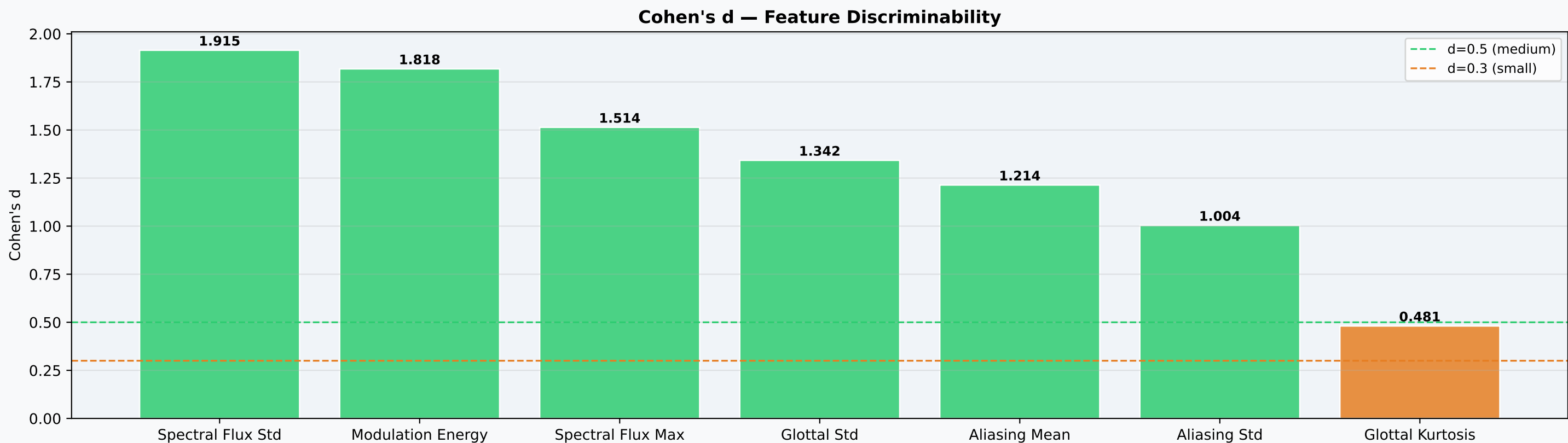
EE 123 Project — Ved, Ching, Erick

Real : 1000 files (source-vctk1k)

Spoof: 1000 files (converted-vctk1k)

Models trained on source vs converted (1000 vs 1000, balanced)

Feature Discriminability — VCTK 1000 Pairs



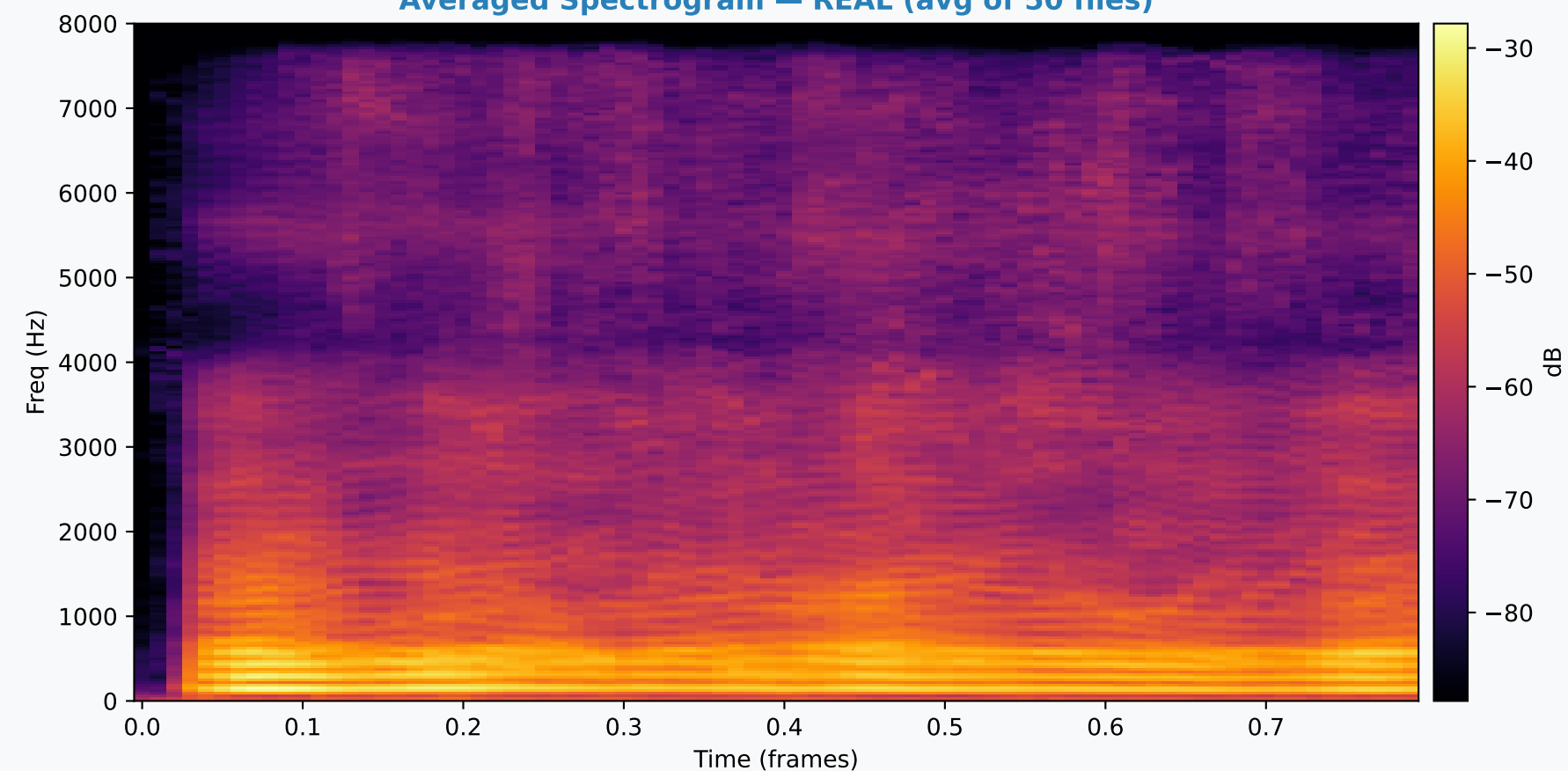
VCTK 1000-Pair Aggregate Feature Statistics

Feature	Real Mean	Real Std	Spoof Mean	Spoof Std	Diff%	Cohen's d	Real>Spoof (out of 1000 pairs)	Consist%	Keep?
Spectral Flux Std	189.228	47.841	119.153	19.744	+37.0%	1.915	943/1000	94.3%	✓ KEEP
Modulation Energy	249.936	80.930	138.952	30.066	+44.4%	1.818	927/1000	92.7%	✓ KEEP
Spectral Flux Max	822.462	254.223	520.393	122.480	+36.7%	1.514	880/1000	88.0%	✓ KEEP
Glottal Std	0.023	0.008	0.015	0.004	+35.5%	1.342	829/1000	82.9%	✓ KEEP
Aliasing Mean	0.208	0.108	0.106	0.049	+49.0%	1.214	792/1000	79.2%	✓ KEEP
Aliasing Std	0.299	0.160	0.173	0.077	+42.2%	1.004	756/1000	75.6%	✓ KEEP
Glottal Kurtosis	16.562	7.461	20.699	9.612	-25.0%	0.481	331/1000	33.1%	✗ DROP

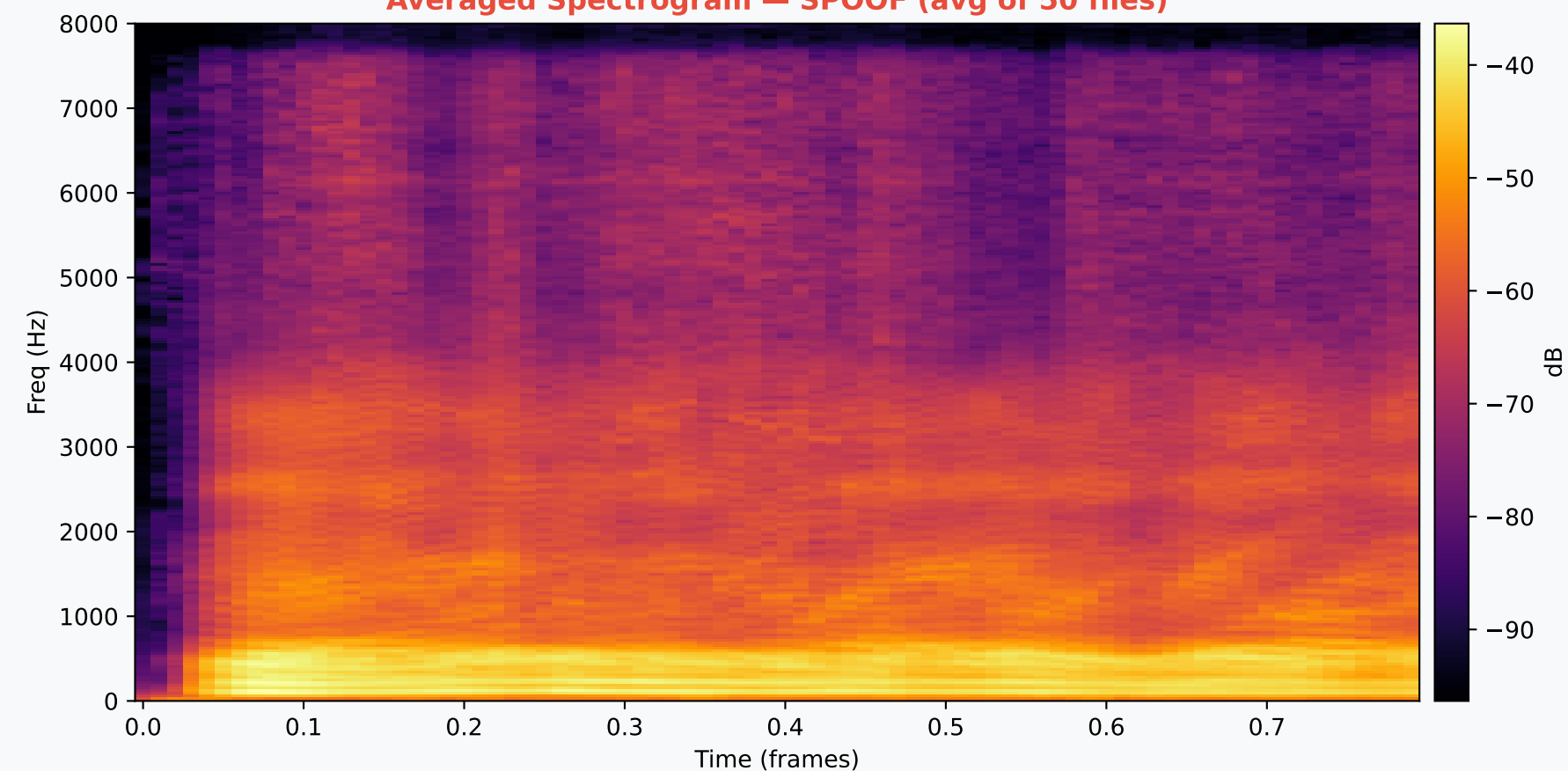
Green = strong ($d>0.5$) | Yellow = moderate ($d>0.3$) | Red = weak/drop

Averaged Signal Analysis — VCTK 1000 Pairs (50 files sampled per class)

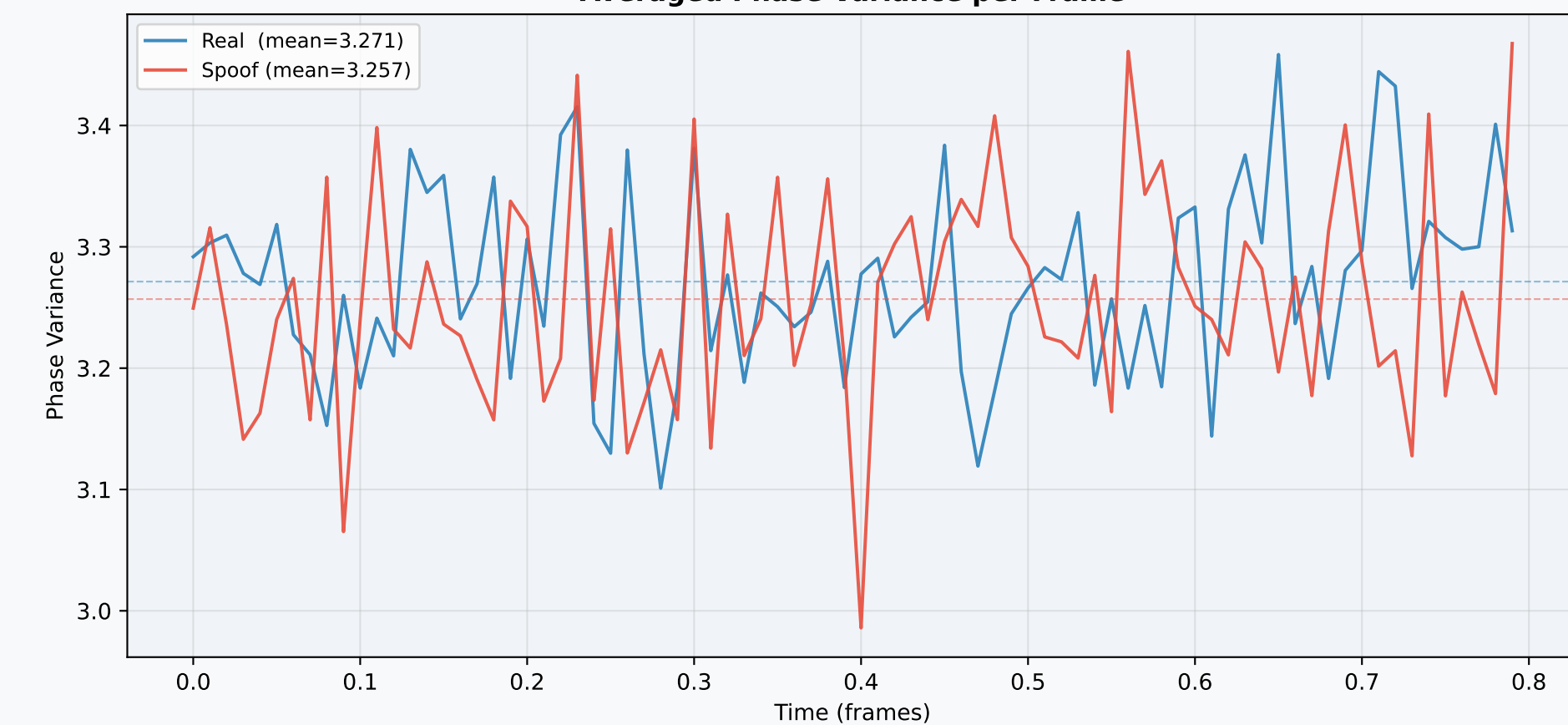
Averaged Spectrogram — REAL (avg of 50 files)



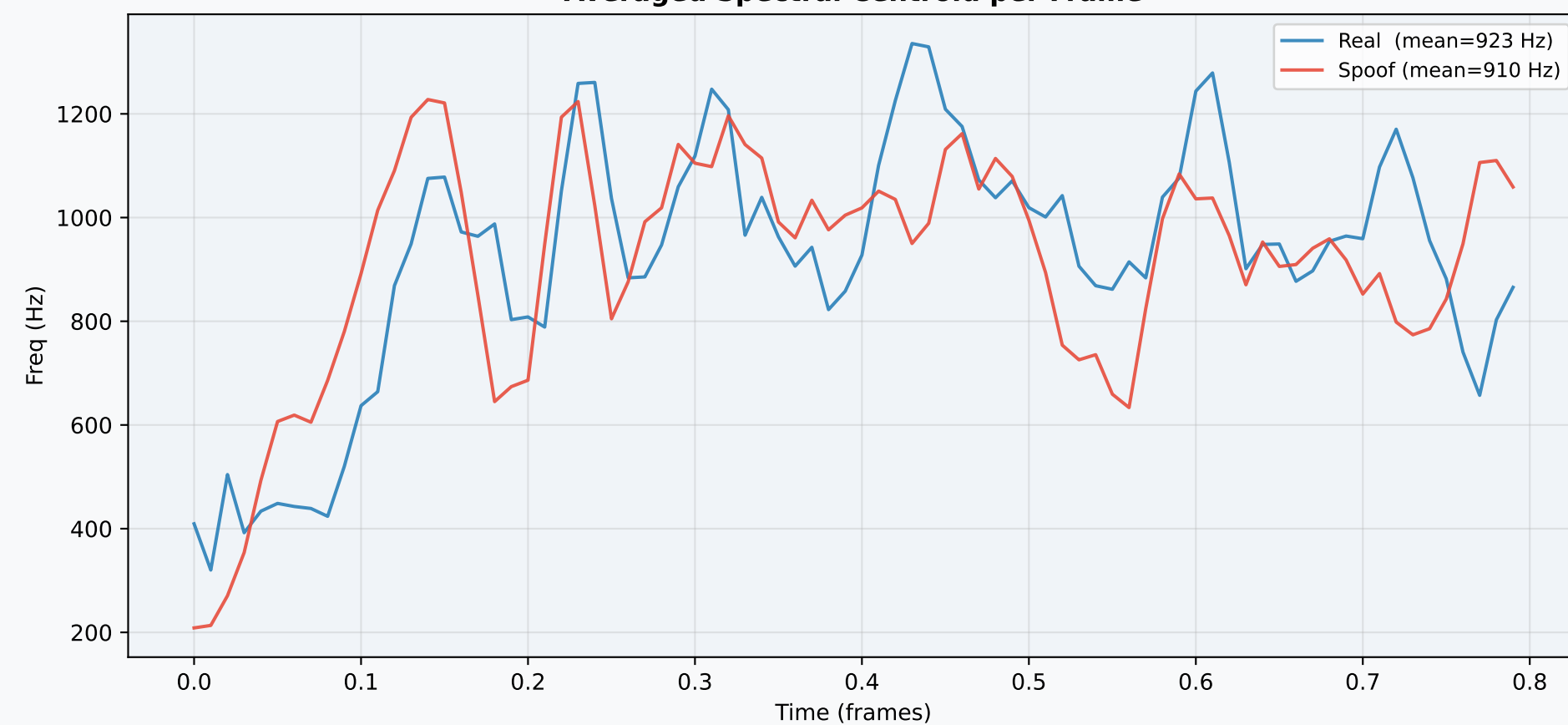
Averaged Spectrogram — SPOOF (avg of 50 files)



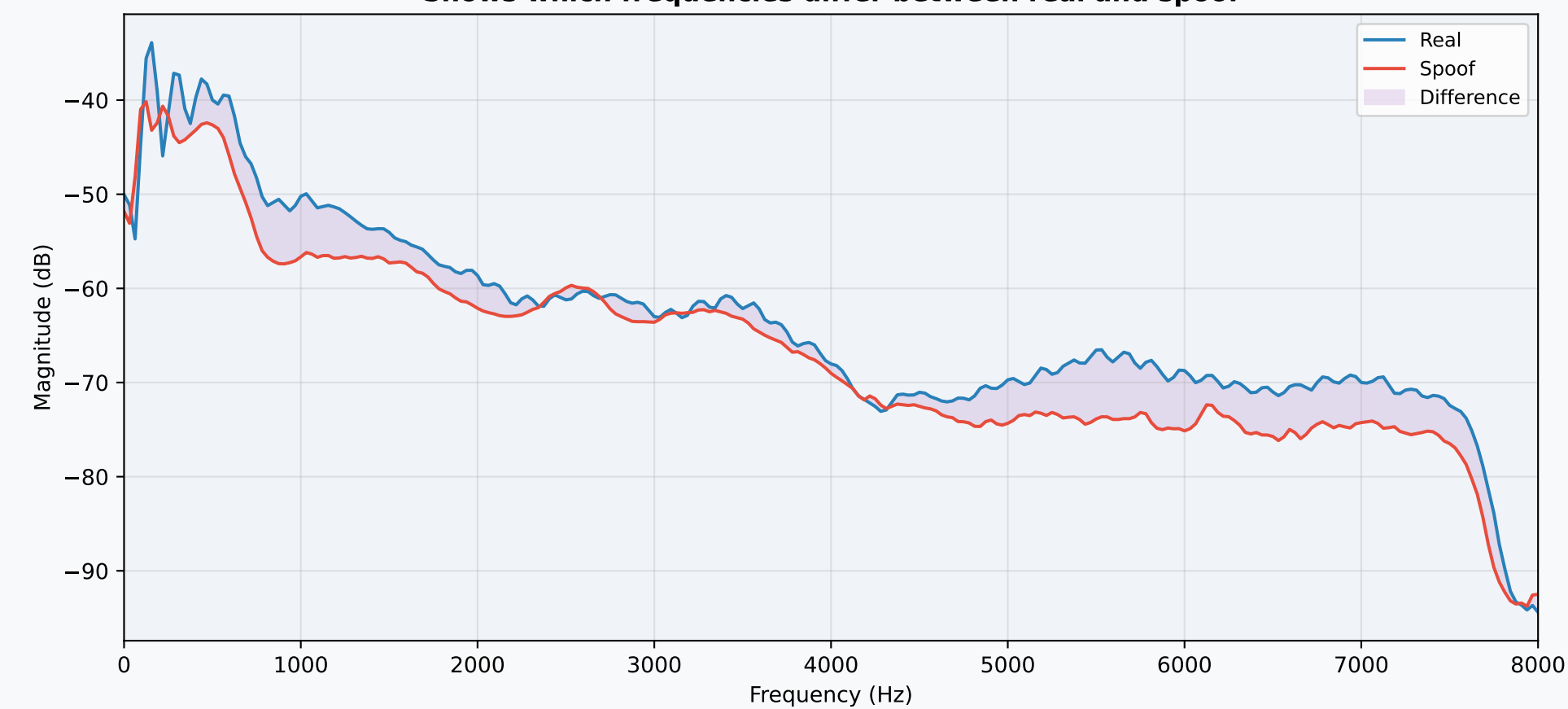
Averaged Phase Variance per Frame



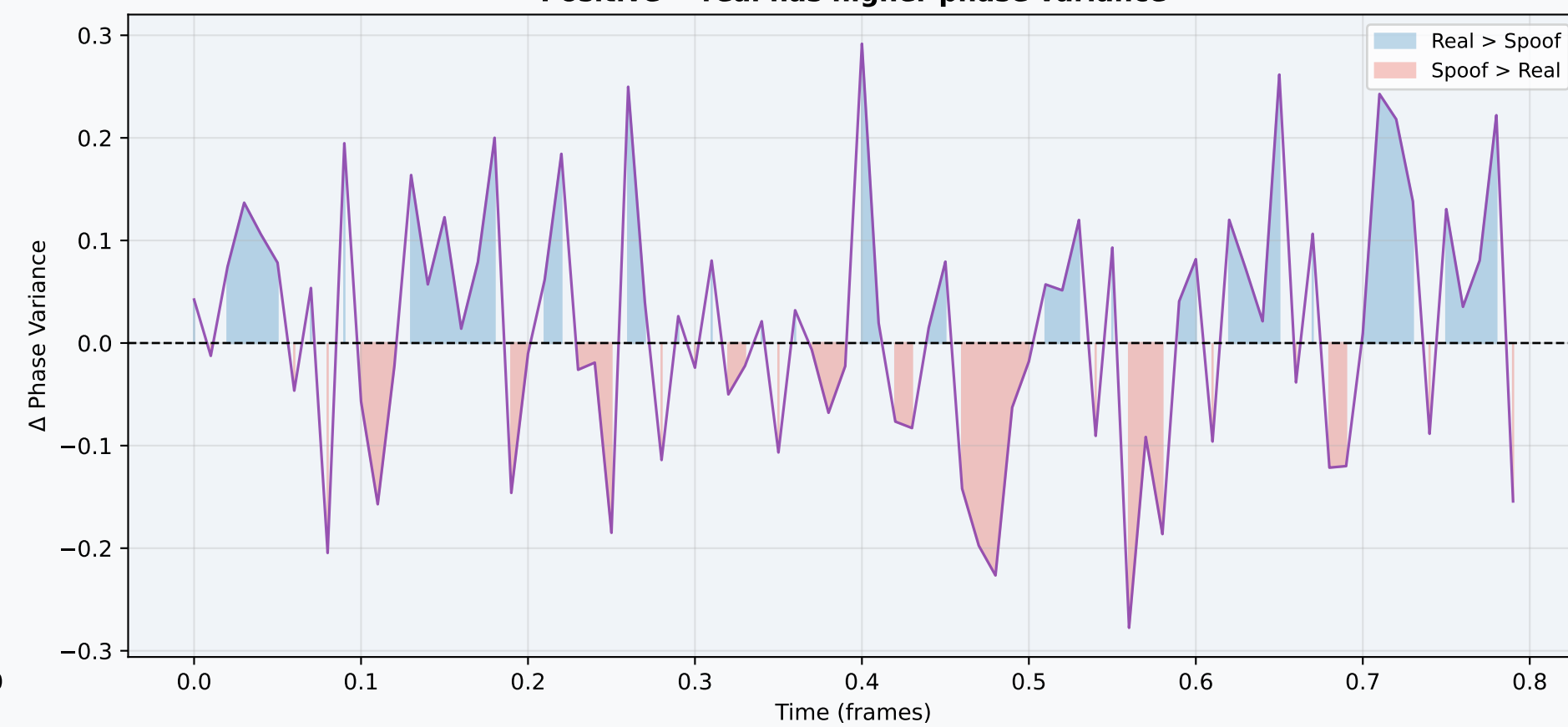
Averaged Spectral Centroid per Frame



Averaged Magnitude Spectrum (dB)
Shows which frequencies differ between real and spoof

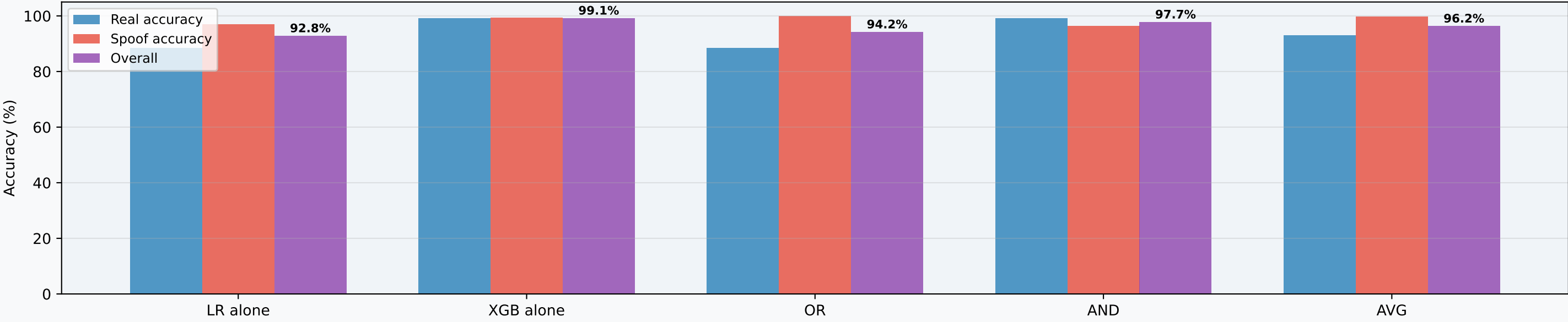


Phase Variance Difference (Real – Spoof)
Positive = real has higher phase variance

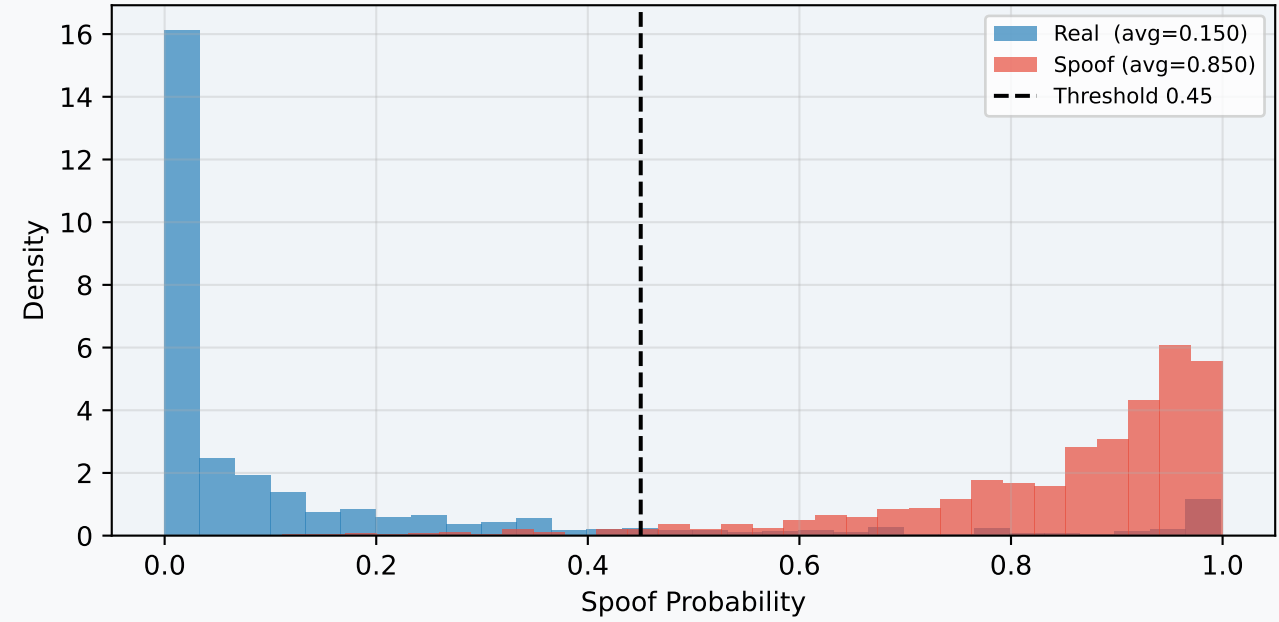


VCTK 1000-Pair — Ensemble Evaluation Results

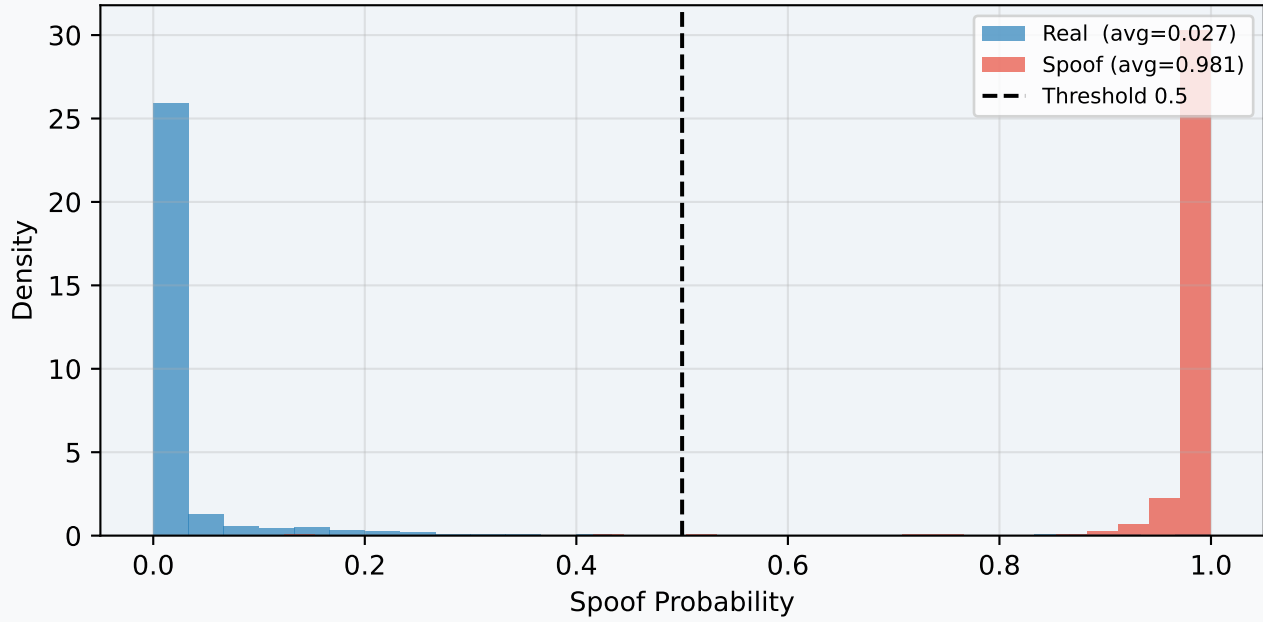
Accuracy by Strategy



LR Probability Distribution



XGBoost Probability Distribution



Raw Count Summary

Strategy	Real Correct	Real Total	Real%	Spoof Correct	Spoof Total	Spoof%	Overall Correct	Overall Total	Overall%
LR alone	885	1000	88.5%	970	1000	97.0%	1855	2000	92.8%
XGB alone	990	1000	99.0%	992	1000	99.2%	1982	2000	99.1%
OR	885	1000	88.5%	998	1000	99.8%	1883	2000	94.2%
AND	990	1000	99.0%	964	1000	96.4%	1954	2000	97.7%
AVG	929	1000	92.9%	996	1000	99.6%	1925	2000	96.2%